

ON BOARD TRAINING RECORD BOOK

FOR
OFFICERS IN CHARGE OF AN
ENGINEERING WATCH (ENGINE CADETS)

Based on the competence
requirements of the 2010
amendments to the
IMO STCW Convention



International Shipping Federation



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FOR _____
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Based on the competence requirements of the 2010 amendments to the IMO STCW Convention

Name

Home Address

.....

.....

.....

Date Training Started



International Shipping Federation

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THIRD EDITION

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The assistance of Stephen E Chapman, FNI is gratefully acknowledged.

1st Edition 1993
2nd Edition 1999
3rd Edition 2013

Published by

Marisec Publications
38 St Mary Axe
London, EC3A 8BH

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www.ics-shipping.org

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ON BOARD TRAINING RECORD BOOK FOR OFFICERS IN CHARGE OF AN ENGINEERING WATCH (ENGINE CADETS)

INTRODUCTION

In 2010, the IMO Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW) was revised and updated. This Training Record Book takes full account of these new requirements for engineer cadets, including new competence standards for leadership and team working, and proactive measures to protect the marine environment.

The revised STCW Convention continues to place emphasis on assessment of the outcome of training, i.e. the ability of seafarers to perform their duties competently. In particular, the 2010 Convention requires that a cadet's seagoing service must be properly structured and recorded in a training record book approved by the maritime administration responsible for issuing certificates of competence. The footnotes of the amended STCW text specifically refer to ISF training record books as an example of such documentation.

The STCW Code, which contains the detailed requirements of the revised STCW Convention, sets out new uniform standards for the attainment of competences in the various maritime skills required to qualify as a watchkeeping officer. The STCW Code also stipulates criteria by which a trainee's attainment of these competences should be assessed by designated on board training officers.

The tasks contained in this Record Book have been carefully designed to help ensure that trainees meet the requirements for certification stipulated by the STCW competences¹ and that as far as possible the officers supervising their training use evaluation based on Table A-III/1 of the STCW Code. However, the tasks have been arranged with on board training in mind. The training tasks and associated criteria are, in many instances, presented in more detail than in the text of the Convention. This is to help ensure that trainees make the best use of their seagoing service and to help officers supervising trainees make a more objective evaluation of whether they are indeed competent.

Normally completion of this Book will not itself constitute an official assessment of competence.² This is the task of government appointed examiners. However, completion of the ISF On Board Training Record Book should provide sufficient documentary evidence that a trainee has completed a properly structured on board training programme and demonstrated competence in the skills required by the amended STCW Convention.

¹ Additional guidance on the competence-based approach to training is given in the ISF Guidelines to the IMO STCW Convention and Code, 2010 (3rd Edition)

² Certain administrations, however, conduct official assessments of seafarers' competence on board. This Book has been written so that it can also be used for official assessment purposes by government approved 'in-service' assessors



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SECTION 1 GUIDE TO COMPLETION

FOR THE ATTENTION OF MASTERS, CHIEF ENGINEERS, DESIGNATED ON BOARD TRAINING OFFICERS AND CADETS

Purpose of the Record Book

The purpose of this Record Book is to help ensure that cadets follow a structured programme of training and make best use of their time at sea. In so doing they will gain the practical training and experience necessary to become competent watchkeeping engineer officers, in accordance with the STCW Convention, as amended in 2010. It is therefore important that this guidance is carefully followed.

Given that this Training Record Book will be submitted to government appointed examiners³ (and may also be made available to instructors at maritime training colleges) proper use and completion of this Book is essential. It should be subject to close scrutiny by the chief engineers of the ships on which the cadet serves, by the cadet's designated on board training officers and the shipping company.

³ See footnote on page 7 below

On receipt of this Book:

The trainee should complete the information required on the following pages, including details of Basic Training received in accordance with the STCW Convention. The trainee will then be personally responsible for the safe keeping of this Record Book throughout training.

- Section 3, concerning details of mandatory Safety Familiarisation (page 17) and mandatory Shipboard Familiarisation (page 18), should be completed immediately after the trainee joins each ship. An officer should sign to signify that mandatory familiarisation as required by the STCW Convention has been undertaken.

As soon as possible after joining each ship:

- The trainee should complete Section 4 (page 20) concerning the technical details of the vessel. The chief engineer and the designated training officer on board each ship should provide an opportunity for this exercise to be undertaken.
- The designated on board training officer appointed by the chief engineer should inspect this Book in order to check progress already made. A plan should be made to tackle the competences that still need to be demonstrated.

Throughout the cadet's seagoing service:

- Section 7, which contains a list of on board training tasks, should be progressively completed. Additional guidance on recording progress is given at the start of Section 6 (page 28). Special attention should also be given to the completion of Section 5 (page 26) concerning safety at work.
- The Book should be submitted to the chief engineer⁴ for inspection every month and at the end of each voyage. The chief engineer's comments should be recorded, dated and stamped on page 13. Comments should only relate to the trainee's competence and practical progress.
- The Book should be submitted to the designated on board training officer on joining each vessel – and then, so far as the voyage pattern allows, every week. Comments should be recorded on page 11.
- A precise record should be kept of the trainee's seagoing service including time spent on engine room watchkeeping duties.

Important note

The STCW Convention requires that any person conducting on board training shall do so only when it will not adversely affect the normal operation of the ship and time can be dedicated to the training and any evaluation of competence.

⁴ Some administrations may require government approved 'in-service' assessors to complete sections of this Book as part of the official assessment of competence that may take place on board ship

SECTION 2 SUMMARY RECORD OF PROGRESS

PARTICULARS OF CADET to be completed by the trainee in **BLOCK CAPITALS**

Name in Full

Seafarer's Book No. Date of Birth

Home Address

.....

Change of Address (if applicable)

.....

Sponsoring Company

Address

.....

Cadet Agreement Date Started Date Finished

Change of Sponsoring Company (if applicable)

Address

.....

Date of Change Date Finished

PHOTO

TRAINING PROGRAMME as applicable

COLLEGE PHASES:		
	From:	To:
	From:	To:
	From:	To:
	From:	To:
SEA PHASES:		
	From:	To:
	From:	To:
	From:	To:
WORKSHOP OR OCCUPATIONAL TRAINING:		
	From:	To:
	From:	To:
	From:	To:

BASIC TRAINING as required by Section A-VI/1 paragraph 2 of the STCW Code

As part of your pre-sea training you should have completed Basic Training or instruction as listed. Enter details of this training or instruction below.

	Date	Location	Document Number
Personal Survival Techniques			
Fire Prevention and Fire-Fighting			
Elementary First Aid			
Personal Safety and Social Responsibilities			

SHIPBOARD RECORD OF SERVICE

Ship	IMO Number	Dates		Time Spent on Engine Room Watchkeeping Functions		Voyage Total - Seagoing Service	
		Sign on	Sign off	Months	Days	Months	Days
Total Service							

DESIGNATED TRAINING OFFICER'S REVIEW OF TRAINING PROGRESS

This table should be completed at least once a week or at such intervals as the trading of the vessel allows.
Comments should only relate to the cadet's practical progress and competence and should NOT refer to character.

Ship	Comments	Name in BLOCK CAPITALS	Initials	Date

CHIEF ENGINEER'S MONTHLY INSPECTION OF RECORD BOOK

Comments should only relate to the cadet's practical progress and competence and should NOT refer to character.

Ship	Comments	Chief Engineer's Name in BLOCK CAPITALS	Chief Engineer's Initials	Date	Ship's Official Stamp

COMPANY'S INSPECTION OF RECORD BOOK

Comments should only relate to the cadet's practical progress and competence and should NOT refer to character.

Comments	Name in BLOCK CAPITALS	Initials	Date

SECTION 3 MANDATORY SAFETY AND SHIPBOARD FAMILIARISATION

SAFETY FAMILIARISATION as required by Section A-VI/1 paragraph 1 of the STCW Code

Before being assigned to shipboard duties all seafarers must receive basic safety familiarisation to know what to do in an emergency. The chief engineer or responsible officer on each ship should sign and date below to signify that you have received training or instruction to be able to carry out the following tasks or duties.

Ship's Name						
Task/Duty	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date
Be able to: Communicate with other persons on board on elementary safety matters						
Understand safety information symbols, signs and alarm signals						
Know what to do if: A person falls overboard Fire or smoke is detected The fire or abandon ship alarm is sounded						
Be able to: Identify muster and embarkation stations and emergency escape routes						
Locate and don life jackets and survival suits						
Raise the alarm and have a basic knowledge of the use of portable fire extinguishers						
Take immediate action upon encountering an accident or other medical emergency before seeking further medical assistance on board						
Close and open the fire, weathertight and watertight doors fitted in the particular ship, other than those for hull openings						

SHIPBOARD FAMILIARISATION as required by Regulation I/14 of the STCW Convention

You will be given a period of time during which you will have an opportunity to become acquainted with the equipment you will be using, and specific watchkeeping, safety, environmental and emergency procedures and arrangements required to perform your duties. The location of safety and emergency equipment varies from ship to ship. To be sure that you are familiar with your duties and all ship arrangements, installations, equipment procedures and ship characteristics that are relevant to your routine or emergency duties, you must complete the following tasks or duties as soon as possible on joining your ship.

Ship's Name						
Task/Duty	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date
Watchkeeping procedures and arrangements:						
Have knowledge of engine room (ER) and other work areas						
Have knowledge of main and auxiliary engines and other engine room equipment and displays						
Operate, under supervision, equipment, plant and machinery to be used in routine duties						
Safety and emergency procedures:						
Read and demonstrate an understanding of your Company's Fire and Safety Regulations						
Demonstrate recognition of the engine room and general alarm signals for: FIRE EMERGENCY ABANDON SHIP ENGINE ROOM CO₂ RELEASE						
Locate engine room first aid equipment						
Locate Emergency Escape Breathing Devices (EEBDs) for machinery space and accommodation						
Locate fire-fighting equipment: alarm activating points, alarm bells, extinguishers, hydrants, breathing apparatus, fire-fighter's outfits and hoses						
Locate rocket line throwing apparatus						
Locate distress rockets, flares and other pyrotechnics						
Locate EPIRB, SART and portable radios for use in emergency						
Locate CO ₂ bottle room, and release points and control valves for machinery spaces, engine room, pump rooms, cargo tanks and holds						

SHIPBOARD FAMILIARISATION as required by Regulation I/14 of the STCW Convention (continued)

Ship's Name						
Task/Duty	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date	Officer's Initials/Date
Safety and emergency procedures (continued): Locate and understand the operation of the emergency deck stops for main engines, fire flaps, ventilation, fuel oil valve and other emergency stop valves						
Locate and understand the operation of the emergency fire pump, emergency generator and emergency compressor						
Environmental protection: Get acquainted with: The procedure for handling garbage, rubbish and other wastes						
Handling of oily bilge water and oil wastes						

BOAT AND MUSTER STATIONS

Insert Boat and Fire Muster Stations and other details in the appropriate space. Ask the master to sign in the space provided.

Ship's Name						
Boat Muster Station						
Fire Muster Station						
Master's Name BLOCK CAPITALS						
Master's Signature						
Date						

SECTION 4 PARTICULARS OF SHIPS

It is an essential feature of your training that you obtain knowledge of the ships on which you serve. To assist you in meeting this important requirement the following particulars are to be recorded during the time spent on each ship. Questions on this subject, with particular reference to your last ship, are likely to be put to you during an oral examination and assessment for your certificate of competency.

FIRST SHIP

mv/ss	IMO Number	Call Sign
Dimensions and Capacities	Life-Saving Equipment	Steering Gear
Length overall m	Lifeboats (no.)	Type.....
Breadth m	Life rafts (no.)	Cargo Handling Gear
Depth m	Capacity per boat (persons)	Derricks/cranes (no. and SWL) tonnes
Summer draft m	Capacity per life raft (persons)	Winches (types) tonnes
Summer freeboard m	Survival Suits (no./type)	Other cargo equipment
Gross tonnage t	Emergency Escape Breathing Devices (EEBDs) (no./type)	Ballast tanks (no.).....
Deadweight..... t	Fire-Fighting Equipment	Cargo tanks (no.).....
Light displacement..... t	Fire extinguishers (no. and capacity)	Cargo pumps (no.).....
Grain/liquid capacity m ³	Types: Water litres Foam litres	Pipelines (sizes).....
Main Engines	Dry powder kg CO ₂ kg	(type and rating) tonnes/hour
Engine (make/type)	Fire hoses (no. and size) mm	Anchors
Stroke Bore	Breathing apparatus (make)	Port weight tonnes
Output bhp/kW @ rpm	ER fixed fire-fighting system (type)	Starboard weight tonnes
Turbo charger	Other fixed fire-fighting system(s) (type)	Spare weight tonnes
Reduction gears type	Auxiliaries	Cable (diameter) mm
Type of waste heat recovery	Generators (type/make)	Length shackles
Engine fuel type Cons t/d	Output	Windlass (make/type)
Viscosity cSt at °C	Fuel type Cons t/d	
Auxiliary boilers (type and no.)	Purifiers (type/make/capacity)	
Make	LO HFO MDO	
Working pressure kg/m ² or bar		

SECOND SHIP

mv/ss	IMO Number	Call Sign
Dimensions and Capacities	Life-Saving Equipment	Steering Gear
Length overall m	Lifeboats (no.)	Type.....
Breadth m	Life rafts (no.)	Cargo Handling Gear
Depth m	Capacity per boat (persons)	Derricks/cranes (no. and SWL) tonnes
Summer draft m	Capacity per life raft (persons)	Winches (types) tonnes
Summer freeboard m	Survival Suits (no./type)	Other cargo equipment.....
Gross tonnage t	Emergency Escape Breathing Devices (EEBDs)	
Deadweight..... t	(no./type)	Ballast tanks (no.).....
Light displacement..... t	Fire-Fighting Equipment	Cargo tanks (no.).....
Grain/liquid capacity m ³	Fire extinguishers (no. and capacity)	Cargo pumps (no.).....
Main Engines	Types: Water litres Foam litres	Pipelines (sizes).....
Engine (make/type)	Dry powder kg CO ₂ kg	(type and rating) tonnes/hour
Stroke Bore	Fire hoses (no. and size) mm	Anchors
Output bhp/kW @ rpm	Breathing apparatus (make)	Port weight tonnes
Turbo charger	ER fixed fire-fighting system (type)	Starboard weight tonnes
Reduction gears type	Other fixed fire-fighting system(s) (type)	Spare weight tonnes
Type of waste heat recovery	Auxiliaries	Cable (diameter) mm
Engine fuel type Cons t/d	Generators (type/make)	Length shackles
Viscosity cSt at °C	Output	Windlass (make/type)
Auxiliary boilers (type and no.)	Fuel type Cons t/d	
Make	Purifiers (type/make/capacity)	
Working pressure kg/m ² or bar	LO HFO MDO	

THIRD SHIP

mv/ss	IMO Number	Call Sign
Dimensions and Capacities	Life-Saving Equipment	Steering Gear
Length overall m	Lifeboats (no.)	Type
Breadth m	Life rafts (no.)	Cargo Handling Gear
Depth m	Capacity per boat (persons)	Derricks/cranes (no. and SWL) tonnes
Summer draft m	Capacity per life raft (persons)	Winches (types) tonnes
Summer freeboard m	Survival Suits (no./type)	Other cargo equipment
Gross tonnage t	Emergency Escape Breathing Devices (EEBDs)	
Deadweight t	(no./type)	Ballast tanks (no.)
Light displacement t	Fire-Fighting Equipment	Cargo tanks (no.)
Grain/liquid capacity m ³	Fire extinguishers (no. and capacity)	Cargo pumps (no.)
Main Engines	Types: Water litres Foam litres	Pipelines (sizes)
Engine (make/type)	Dry powder kg CO ₂ kg	(type and rating) tonnes/hour
Stroke Bore	Fire hoses (no. and size) mm	Anchors
Output bhp/kW @ rpm	Breathing apparatus (make)	Port weight tonnes
Turbo charger	ER fixed fire-fighting system (type)	Starboard weight tonnes
Reduction gears type	Other fixed fire-fighting system(s) (type)	Spare weight tonnes
Type of waste heat recovery	Auxiliaries	Cable (diameter) mm
Engine fuel type Cons t/d	Generators (type/make)	Length shackles
Viscosity cSt at °C	Output	Windlass (make/type)
Auxiliary boilers (type and no.)	Fuel type Cons t/d	
Make	Purifiers (type/make/capacity)	
Working pressure kg/m ² or bar	LO HFO MDO	

FOURTH SHIP

mv/ss	IMO Number	Call Sign
Dimensions and Capacities	Life-Saving Equipment	Steering Gear
Length overall m	Lifeboats (no.)	Type.....
Breadth m	Life rafts (no.)	Cargo Handling Gear
Depth m	Capacity per boat (persons)	Derricks/cranes (no. and SWL) tonnes
Summer draft m	Capacity per life raft (persons)	Winches (types) tonnes
Summer freeboard m	Survival Suits (no./type)	Other cargo equipment.....
Gross tonnage t	Emergency Escape Breathing Devices (EEBDs)	
Deadweight..... t	(no./type)	Ballast tanks (no.).....
Light displacement..... t	Fire-Fighting Equipment	Cargo tanks (no.).....
Grain/liquid capacity m ³	Fire extinguishers (no. and capacity)	Cargo pumps (no.).....
Main Engines	Types: Water litres Foam litres	Pipelines (sizes).....
Engine (make/type)	Dry powder kg CO ₂ kg	(type and rating) tonnes/hour
Stroke Bore	Fire hoses (no. and size) mm	Anchors
Output bhp/kW @ rpm	Breathing apparatus (make)	Port weight tonnes
Turbo charger	ER fixed fire-fighting system (type)	Starboard weight tonnes
Reduction gears type	Other fixed fire-fighting system(s) (type)	Spare weight tonnes
Type of waste heat recovery	Auxiliaries	Cable (diameter) mm
Engine fuel type Cons t/d	Generators (type/make)	Length shackles
Viscosity cSt at °C	Output	Windlass (make/type)
Auxiliary boilers (type and no.)	Fuel type Cons t/d	
Make	Purifiers (type/make/capacity)	
Working pressure kg/m ² or bar	LO HFO MDO	

FIFTH SHIP

mv/ss	IMO Number	Call Sign
Dimensions and Capacities	Life-Saving Equipment	Steering Gear
Length overall m	Lifeboats (no.)	Type.....
Breadth m	Life rafts (no.)	Cargo Handling Gear
Depth m	Capacity per boat (persons)	Derricks/cranes (no. and SWL) tonnes
Summer draft m	Capacity per life raft (persons)	Winches (types) tonnes
Summer freeboard m	Survival Suits (no./type)	Other cargo equipment.....
Gross tonnage t	Emergency Escape Breathing Devices (EEBDs) (no./type)	
Deadweight t	Fire-Fighting Equipment	Ballast tanks (no.).....
Light displacement..... t	Fire extinguishers (no. and capacity)	Cargo tanks (no.).....
Grain/liquid capacity m ³	Types: Water litres Foam litres	Cargo pumps (no.).....
Main Engines	Dry powder kg CO ₂ kg	Pipelines (sizes).....
Engine (make/type)	Fire hoses (no. and size) mm	(type and rating) tonnes/hour
Stroke Bore	Breathing apparatus (make)	Anchors
Output bhp/kW @rpm	ER fixed fire-fighting system (type)	Port weight tonnes
Turbo charger	Other fixed fire-fighting system(s) (type)	Starboard weight tonnes
Reduction gears type	Auxiliaries	Spare weight tonnes
Type of waste heat recovery	Generators (type/make)	Cable (diameter) mm
Engine fuel type Cons t/d	Output	Length shackles
Viscosity cSt at °C	Fuel type Cons t/d	Windlass (make/type)
Auxiliary boilers (type and no.)	Purifiers (type/make/capacity)	
Make	LO HFO MDO	
Working pressure kg/m ² or bar		

SIXTH SHIP

mv/ss	IMO Number	Call Sign
Dimensions and Capacities	Life-Saving Equipment	Steering Gear
Length overall m	Lifeboats (no.)	Type
Breadth m	Life rafts (no.)	Cargo Handling Gear
Depth m	Capacity per boat (persons)	Derricks/cranes (no. and SWL) tonnes
Summer draft m	Capacity per life raft (persons)	Winches (types) tonnes
Summer freeboard m	Survival Suits (no./type)	Other cargo equipment
Gross tonnage t	Emergency Escape Breathing Devices (EEBDs)	
Deadweight t	(no./type)	Ballast tanks (no.)
Light displacement t	Fire-Fighting Equipment	Cargo tanks (no.)
Grain/liquid capacity m ³	Fire extinguishers (no. and capacity)	Cargo pumps (no.)
Main Engines	Types: Water litres Foam litres	Pipelines (sizes)
Engine (make/type)	Dry powder kg CO ₂ kg	(type and rating) tonnes/hour
Stroke Bore	Fire hoses (no. and size) mm	Anchors
Output bhp/kW @ rpm	Breathing apparatus (make)	Port weight tonnes
Turbo charger	ER fixed fire-fighting system (type)	Starboard weight tonnes
Reduction gears type	Other fixed fire-fighting system(s) (type)	Spare weight tonnes
Type of waste heat recovery	Auxiliaries	Cable (diameter) mm
Engine fuel type Cons t/d	Generators (type/make)	Length shackles
Viscosity cSt at °C	Output	Windlass (make/type)
Auxiliary boilers (type and no.)	Fuel type Cons t/d	
Make	Purifiers (type/make/capacity)	
Working pressure kg/m ² or bar	LO HFO MDO	

SECTION 5 SAFETY AT WORK

Ships and ships' engine rooms can be dangerous places in which to work. Taking proper precautions will minimise the risks. Whilst the master is responsible for the overall safety of the ship and those on board, individual crew members have a duty to ensure safety in those matters within their own control. All the safeguards and other facilities provided for your safety should be used.

In all engine room tasks there is a safe way and an unsafe way to proceed. Give plenty of thought to what you are doing, keep your eyes and ears open and aim to be a safe engineer. Don't take risks. Follow the correct procedures. Wear suitable clothing and protective footwear, and always use the personal protective equipment provided, for example hard hat, hearing protection, goggles, gloves etc.

Ref No	Training		Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
	Competence: Maintain safe operations				
	Application of safe working practices on board		Operations, maintenance and repairs are planned and carried out in accordance with safety rules and procedures		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)	Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Describe the system of permits to work on board				
.2	List the items to be checked in a work permit				
.3	List the items to be checked in a hot work permit				
.4	Describe an enclosed space				
.5	Describe the procedures to enter an enclosed space				
.6	Explain the use of gas analysis instruments to be used prior to entering: Fuel oil tanks				
.7	Ballast tanks				
.8	Void spaces				

.9	Describe the procedure adopted on finding someone overcome as a result of: Electric shock					
.10	Gassing incident in an enclosed space					
.11	Describe special safety precautions in dry dock					
.12	Demonstrate an understanding of safe working practices for use of welding and cutting equipment					

SECTION 6 INFORMATION ON TRAINING TASKS AND COMPETENCES TO BE ACHIEVED

This section of your Record Book gives details of the training tasks that you should follow to make best use of your time at sea. You will see that each page lists the tasks or duties that you should undertake. Completion of these will lead to meeting the competences.

A senior officer should review your progress and indicate, with initials and date in the light **green** box on the right hand side of the page, that your performance is considered to meet the Criteria for Evaluation and that competence has been demonstrated in that element. The officer may offer advice on areas in which improvement is necessary. The competences required to qualify as a watchkeeping officer as tabulated in the STCW Code are listed below. This Section is organised as follows:

COMPETENCES FOR OFFICERS IN CHARGE OF AN ENGINEERING WATCH

Marine Engineering at the Operational Level

1. Maintain a safe engineering watch (page 30)
2. Use English in written and oral form (page 34)
3. Use internal communication systems (page 35)
4. Operate main and auxiliary machinery and associated control systems (page 36)
5. Operate fuel, lubrication, ballast and other pumping systems and associated control systems (page 40)

Electrical, Electronic and Control Engineering at the Operational Level

6. Operate electrical, electronic and control systems (page 42)
7. Maintenance and repair of electrical and electronic equipment (page 46)

Maintenance and Repair at the Operational Level

8. Appropriate use of hand tools, machine tools and measuring instruments for fabrication and repair on board (page 53)
9. Maintenance and repair of shipboard machinery and equipment (page 59)

Controlling the Operation of the Ship and Care for Persons On Board at the Operational Level

10. Application of leadership and teamworking skills (page 67)⁵
11. Ensure compliance with pollution prevention requirements (page 68)
12. Maintain seaworthiness of the ship (page 71)
13. Prevent, control and fight fires on board (page 72)
14. Operate life-saving appliances (page 75)
15. Apply medical first aid on board ship (page 78)
16. Monitor compliance with legislative requirements (page 79)

⁵ For the purpose of organising this book, this competence has been listed first for the function Controlling the Operation of the Ship and Care for Persons On Board at the Operational Level. The STCW competence 'Contribute to the safety of personnel and ship' has been incorporated into competences 12-16.

EXAMPLE OF HOW TO COMPLETE THE LIST OF TRAINING TASKS AND COMPETENCES ACHIEVED

Ref No	Training	Criteria for Evaluation		Competence Demonstrated	
				Designated Training Officer/In Service Assessor (Initials/Date)	
1.	Competence: Maintain a safe engineering watch				
1.5	Complete the engine room log book and other records	All significant readings, movements and activities related to the engineering systems are properly recorded		CM	20/11/11
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	
		Task Completed Supervising Officer/ Instructor (Initials/Date)		Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Complete the engine room log book and record books	AB	9/9/11	Write clearly in the columns and spaces provided Take more care with readings and calculations	CM 16/9/11
.2	Record the complete engine movements in the log during periods of manoeuvring	CM	10/10/11		
.3	Evaluate record entries in the Alarm Record Book	CM	20/9/11		CM 16/11/11

- The competences marked in **green** are directly relevant to the competences required by the 2010 STCW Convention.
- The primary tasks are sub divided into training tasks or duties on the left hand side of the page. The trainee should complete as many of these training tasks as possible. However, in view of the likely equipment, cargo or voyage pattern of the ship, **it is not expected that all these tasks will be completed before the trainee is considered competent in the primary tasks**. It should be noted that some of the skills and knowledge that underpin the competences may well have been obtained during shore based training.
- Space is provided to record completion of each training task twice by the supervising officer. The officer supervising the trainee does not necessarily have to be the designated training officer. This does not mean that each task must be completed twice if, in the opinion of the officer, once is considered sufficient.
- Before 'competence demonstrated' is recorded for the competences marked in green, the chief engineer or designated on board training officer may record any appropriate advice about areas for improvement. A large blank space for this purpose is provided beneath the criteria for evaluation.
- As competence in these primary tasks is demonstrated, the appropriate light green boxes next to the criteria for evaluation, on the far right hand side of the page, should be signed and dated (day, month and year) by the chief engineer or designated training officer on board the ship to attest that competence has been demonstrated.
- A trainee's attainment of the competences marked should only be recorded as 'competence demonstrated' when the chief engineer or designated training officer is indeed satisfied that the trainee can perform the duty without supervision.
- When recording 'competence demonstrated', account should be taken of the criteria for evaluation contained on the right hand side of the page, as well as the best practices of seafarers and safe working practice.

SECTION 7 TASKS FOR OFFICERS IN CHARGE OF AN ENGINEERING WATCH

FUNCTION: MARINE ENGINEERING AT THE OPERATIONAL LEVEL

Ref No	Training			Criteria for Evaluation	Competence Demonstrated	
1.	Competence: Maintain a safe engineering watch				Designated Training Officer/In Service Assessor (Initials/Date)	
1.1	Relieve and hand over the watch			The duties are carried out in accordance with accepted principles, procedures and ship specific instructions. Communication is clearly and unambiguously given and received		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Follow the correct procedure for handing over a watch: At sea					
.2	In port					
.3	Follow the correct procedures for taking over and accepting a watch: At sea					
.4	In port					
1.2	Conduct the watch			The frequency and extent of monitoring machinery, equipment and systems conform to manufacturers' recommendations and accepted principles and procedures and are sufficient to deal with common operational errors and fault conditions. Questionable decisions and/or actions result in appropriate challenge and response		
.1	Assist with the duties of an engineer officer on: Seagoing watches					
.2	Port watches					
.3	Anchor watches					
.4	Under supervision, carry out all routine watchkeeping duties, checking the correct functioning of all automatic control and monitoring systems					

.5	Make adjustments as found necessary					
.6	Perform routine checks in machinery space for correct water levels					
.7	Blow down main engine scavenge drains					
.8	Ensure that compressed air automatic drains are functioning correctly					
.9	Check sheathing on high-pressure fuel pipes					
.10	Clean air side of the turbo charger					
.11	Carry out boiler water tests and corrective treatment					
.12	Check returns from heating coils and other possible sources of contaminated feedwater					
.13	Check the correct operation of the boiler including water level and burner					
.14	Carry out a soot-blowing procedure					
.15	Check all air receiver drains					
.16	Assist on the bridge during manoeuvring operations: Entering port					
.17	Leaving port					
.18	Understand that effective watchkeeping involves managing watch duties, including supervision, as well as maintaining the safe operation of propulsion plant and other machinery					
1.3	Response to black-out and emergency situations			<i>Immediate actions are executed in accordance with laid down procedures, and due regard paid to the actual situation. Effective leadership behaviours are demonstrated</i>		
.1	Take corrective action during emergency drills: Fire drill					
.2	Abandon ship drill					
.3	Black out drill					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated	
1.	Competence: Maintain a safe engineering watch				Designated Training Officer/In Service Assessor (Initials/Date)	
1.3	Response to black-out and emergency situations (continued)			Immediate actions are executed in accordance with laid down procedures, and due regard paid to the actual situation. Effective leadership behaviours are demonstrated		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.4	Assist with/demonstrate use of main engine local control and emergency manoeuvring					
.5	Assist with/demonstrate procedure for returning main engine to normal running					
.6	Demonstrate in a drill, emergency running and manoeuvring procedures					
.7	Demonstrate knowledge of emergency steering gear operation					
.8	Demonstrate a knowledge of how to reset machinery following failure and how to restart plant					
.9	State the priorities for restoring services					
.10	Demonstrate knowledge of first start arrangements					
1.4	Change-over of remote-automatic and local control systems			The operations are decisively carried out and in accordance with procedures stated. Questionable decisions and/or actions result in appropriate challenge and response		
.1	Change over to the stand by system for: Main engines					
.2	Generators					
.3	Main engine system pumps					
.4	Steering gear					
.5	Prepare for stand by engines					

1.5	Complete the engine room log book and other records			<i>All significant readings, movements and activities related to the engineering systems are properly recorded</i>		
.1	Complete the engine room log book and record books					
.2	Record the complete engine movements in the log during periods of manoeuvring					
.3	Evaluate record entries in the Alarm Record Book					
.4	Observe and note performance and condition of machinery using condition monitoring equipment, where appropriate					
.5	Observe and note normal operating temperatures/ pressures					
.6	Demonstrate a knowledge of and understand the purpose of the Alarm Record Book					
1.6	Knowledge of engine room resource management principles			<i>Resources are allocated and assigned as needed in correct priority to perform necessary tasks</i>		
.1	Set realistic plans for allocation and use of engine room resources					
.2	Plan tasks to achieve timely outcome					
.3	Understand that to achieve a goal or an objective the plan must be specific with sufficient detail					
.4	Demonstrate collection and interpretation of management data to assess task progress					
.5	Lead progress review with team members to ensure task is attainable within the plan set					
.6	Lead task review on completion giving credit where due and noting areas where things may be done differently on another occasion					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
2.	Competence: Use English in written and oral form					
2.1	Use English engineering publications, operational manuals and fault finding instructions			<i>The publications and manuals relevant to the engineering duties are correctly interpreted</i>		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	List English language publications or manuals used:					
.2	If appropriate, assist with completion of ship's Planned Maintenance System records in English					
2.2	Communicate with others in English language, as appropriate			<i>Communications are clear and understood</i>		
.1	Demonstrate correct use of terms used in the engine room and names of machinery, equipment and tools					
.2	Give and take orders in English concerning: Routine operations					
.3	Emergency drills					
.4	Ensure that others have understood orders correctly					
.5	Demonstrate an ability to communicate instructions effectively in the English language to a multi-lingual crew					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
3.	Competence: Use internal communication systems					
3.1	Operation of all internal communication systems on board			Transmission and reception of messages are consistently successful. Communication records are complete, accurate and comply with statutory requirements		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Demonstrate operation of the ship's internal phone system					
.2	Use internal message system to send and receive information or instructions					
.3	Understand communication is a two-way exchange and demonstrate this in practice: Steering gear to engine room					
.4	Steering gear to bridge					
.5	Demonstrate correct station ID procedure when using hand held transceivers (portable radios)					
.6	Complete records accurately and in a timely way when recording information received by telephone or hand held transceivers (portable radios)					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
4.	Competence: Operate main and auxiliary machinery and associated control systems					
4.1	Prepare machinery for departure from port			<i>All checks and actions are carried out in accordance with laid down instructions and all auxiliary and control systems are functioning properly. All relevant checks and actions are recorded</i>		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Draw a schematic arrangement of the main engine system, using blocks to indicate the main components					
.2	Prepare and test the steering gear and telegraphs					
.3	Confirm bridge and ER communications					
.4	Check starting air compressor and prepare starting air system					
.5	Prepare main and auxiliary machinery for port departure					
.6	Prepare main and auxiliary machinery for the sea passage					
.7	Demonstrate a knowledge of the use of high level and low level sea suction					
4.2	Operate main and auxiliary machinery			<i>The machinery is operated in accordance with instructions, procedures and safe working practices. All instruments are monitored, necessary adjustments made and required actions carried out and properly recorded</i>		
.1	Sketch, in diagrammatic form, the main systems as appropriate for the ship: Auxiliary engine					
.2	Boiler system					
.3	Start main engine from local and remote control positions					
.4	Carry out post start-up checks of main engine and shafting					
.5	Manually operate main compressor and change over to normal automatic running mode					

.6	Record pressures and temperatures for normal running, and note system valve settings and positions in normal running mode				
.7	Respond to instructions from the bridge and operate the main engine controls during periods of manoeuvring				
.8	Water wash exhaust side main engine turbocharger				
.9	Change local/manual control of machinery and systems to remote/automatic control as appropriate				
.10	Adjust main engine and auxiliary machinery for continuous running				
.11	Report abnormal conditions, making a record of same and note corrective action required				
.12	Prepare for running and operate an evaporator/fresh water generator				
.13	Apply tests and conditioning for purity and potability of fresh water				
.14	Check crankcase oil mist detector and demonstrate action to be taken in case of an alarm				
.15	Check governors				
.16	Take power diagram or readings and calculate mean effective pressure and indicated power				
.17	Carry out routine tests on: Engine cooling water				
.18	Fuel oil				
.19	Lube oil				
.20	Assist with shutting down main engine and auxiliary systems after finishing with engines				
.21	Fill a boiler and raise steam from cold				
.22	Raise the temperature of main engine fuel oil from cold to the correct level				

Ref No	Training			Criteria for Evaluation	Competence Demonstrated	
4.	Competence: Operate main and auxiliary machinery and associated control systems				Designated Training Officer/In Service Assessor (Initials/Date)	
4.2	Operate main and auxiliary machinery (continued)			The machinery is operated in accordance with instructions, procedures and safe working practices. All instruments are monitored, necessary adjustments made and required actions carried out and properly recorded		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.23	Admit steam to a line or system, taking all precautions against thermal and pressure shock and avoiding water hammer					
.24	Check the security of steam pipes and any expansion pieces					
.25	Check that steam traps and drains are functioning					
.26	Close down a steam line, observing procedure for draining					
.27	Check quality of combustion, noting: Smoke from the funnel Clarity around the flame Flame shape, size and colour Excess air, CO ₂ /CO reading Carbon and unburnt fuel deposits					
.28	Check returns from heating coils and other possible sources of contaminated feedwater					
.29	Check the correct functioning of all boiler condition indicators and alarms					
.30	Check that correct boiler water level is maintained					
.31	Demonstrate the correct procedure for blowing down a boiler gauge glass					
.32	Explain the effect of varying the temperature of circulating water					
.33	Start up and operate ship's refrigeration plant					

.34	Make up brine, if appropriate					
.35	Check density of the brine					
.36	Shut down and secure refrigeration/AC plant					
.37	Carry out refrigerant charging procedure					
.38	Carry out leak detection for refrigerant gases					
.39	Replenish driers and filters					
.40	Check pressure tank safety devices					
.41	Put sewage system on line and check correct operation					
.42	Operate waste handling equipment: Incinerator					
.43	Shredder/compactor					
.44	Other (state)					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
5.	Competence: Operate fuel, lubrication, ballast and other pumping systems and associated control systems					
5.1	Plan the operations of auxiliary and piping systems and service plants			Operations are planned and all equipment and control systems checked before operations are executed		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Sketch a line diagram of the oily water separator (OWS) system					
.2	Sketch a line diagram of the ballast water system					
.3	Sketch a line diagram of the engine room bilge water system					
.4	Sketch a line diagram of the hold bilge water system					
.5	Assist with the operation of the OWS					
.6	Demonstrate a knowledge of making correct entries in the Oil Record Book					
.7	Assist with planning ballast water management operations					
.8	Plan and line-up: Ballast water pump					
.9	Bilge pump					
.10	Demonstrate a knowledge of the bilge pump					
.11	Sketch a line diagram of the fuel oil bunker system					
.12	Assist with planning for: Receiving bunkers					
.13	Transfer of fuel from bunker tanks to service tanks					

5.2	Operate the systems for fuel oil, lube oil, ballast, bilge, MARPOL equipment and cargo pumping	The operations are carried out in accordance with rules and procedures to ensure safety of operations and avoid pollution of the marine environment			
.1	Under supervision, transfer fuel from bunkers to service tanks, observing all safety, ship stability and pollution prevention requirements				
.2	Drain water/sludge from settling tanks				
.3	Start, operate and monitor fuel oil purifiers				
.4	Demonstrate a knowledge of Sulphur Emissions Control Areas				
.5	Assist an officer with change over from heavy fuel oil to low viscosity fuel oil and vice versa, where applicable				
.6	Start, operate and monitor lube oil purifiers				
.7	Perform routine checks and top ups to maintain lube oil system tanks at the correct levels				
.8	Assist with loading and discharging cargo tanks, including stripping procedures				
.9	Set up and use an OWS in compliance with MARPOL				
.10	Operate an oil discharge monitor in compliance with MARPOL (oil tankers)				
.11	Use bilge holding tanks				
.12	Observe all pollution prevention requirements				
.13	Open up OWS units, clean all parts and reassemble				
.14	Observing all safety, ship stability and pollution prevention requirements, assist an officer with: Ballasting				
.15	Deballasting				
.16	Pump out hold bilges ensuring that all pollution prevention regulations and requirements are observed				
.17	Demonstrate the emergency arrangements for emptying engine room bilges in the event of flooding				

FUNCTION: ELECTRICAL, ELECTRONIC AND CONTROL ENGINEERING AT THE OPERATIONAL LEVEL

Ref No	Training			Criteria for Evaluation	Competence Demonstrated	
6.	Competence: Operate electrical, electronic and control systems				Designated Training Officer/In Service Assessor (Initials/Date)	
6.1	Basic configuration and operating principles of electrical equipment: Locate and use relevant manuals, drawings, diagrams and instructions for electrical equipment and distribution systems			The instructions and manuals relevant for safe and efficient operations are quickly identified and properly used. Electrical systems can be understood and explained with drawings/instructions		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Explain the difference between a system diagram, a circuit diagram and a wiring diagram					
.2	Demonstrate an ability to use ship's diagrams to identify: Main circuit breakers Emergency switchboard connections Trips (over current, reverse power, low frequency) Transformers Fuses Supply voltages Shore connections Loads to each piece of equipment The types of motors and motor starters					
.3	Demonstrate a knowledge of symbols commonly used on circuit diagrams					
.4	Demonstrate a knowledge of the location of major control and protection devices within the distribution network					
.5	Demonstrate a knowledge of which electrical loads are classed as essential or non-essential, and how essential services are supplied					
.6	Locate shore power connection and state the procedures for connection/disconnection					
6.2	Prepare and start alternators or generators			The operations are planned and carried out in accordance with operating manuals, established rules and procedures to ensure safety of operations		
.1	Assist with pre start-up checks and tests on electrical equipment and control systems					

.2	Prepare for starting in manual and remote modes					
.3	Carry out post start-up checks					
.4	Check that all controls are functioning correctly					
.5	Demonstrate knowledge of trips and how to reset for: Over current					
.6	Reverse power					
.7	Low frequency					
.8	Check exhaust pipes for leakage					
.9	Check efficiency of sheathing on high-pressure fuel pipes and associated leak-off indicators					
6.3	Parallel and change-over alternators or generators			<i>The operations are planned and carried out in accordance with operating manuals, established rules and procedures to ensure safety of operations</i>		
.1	After start-up, run up to speed, use paralleling procedures and put on load, including shaft generators and emergency generators					
.2	Adjust the load share of machines running in parallel					
.3	Remove the load from a machine running in parallel, stop and shut down					
.4	Describe the safety features in the power distribution system which protect alternators in case of a major fault					
6.4	Start electric motors including high voltage installations, where appropriate			<i>The operations are planned and carried out in accordance with operating manuals, established rules and procedures to ensure safety of operations</i>		
.1	Demonstrate an understanding of the starting methods for electric motors					
.2	Start up and operate a high capacity pump					
.3	Demonstrate a knowledge of protective switch gear associated with high voltage installations					
.4	Demonstrate an understanding of the ship's permit to work system concerning electrical equipment					

Ref No	Training	Criteria for Evaluation		Competence Demonstrated	
6.	Competence: Operate electrical, electronic and control systems			Designated Training Officer/In Service Assessor (Initials/Date)	
6.5	Basic configuration and operating principles of electronic equipment: Characteristics of basic electronic circuit elements	The instructions and manuals relevant for safe and efficient operations are quickly identified and properly used. Electronic systems can be understood and explained with drawings/instructions			
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	
.1	Sketch and describe a component providing electronic equipment control				
.2	Assist with routine checks and tests on electronic equipment				
.3	Demonstrate a knowledge of electronic circuit symbols				
.4	Demonstrate a knowledge of the characteristics of basic electronic circuit elements				
6.6	Flow chart for automatic and control systems	The instructions and manuals relevant for safe and efficient operations are quickly identified and properly used. Electronic systems can be understood and explained with drawings/instructions			
.1	Demonstrate a knowledge of process signal symbols and terminology commonly used with control system diagrams				
.2	Sketch a part of the ship's electrical distribution system that uses sequential control circuits				
.3	List other items of equipment that use sequential control circuits				
.4	Demonstrate a knowledge of flowcharts for automatic and control systems for electronic equipment operation				
.5	Assist with routine checks and tests on electronic equipment control systems				

6.7	Functions, characteristics and features of control systems for machinery			The instructions and manuals relevant for safe and efficient operations are quickly identified and properly used. Electronic systems can be understood and explained with drawings/instructions		
.1	Sketch and describe a system of electronic control					
.2	Demonstrate a knowledge of the functions, characteristics and features of the control system for: Main propulsion engine					
.3	Steam boiler					
.4	Steering gear					
6.8	Basic configuration and operating principles of electrical and electronic control systems: Automatic control methodologies and characteristics			The instructions and manuals relevant for safe and efficient operations are quickly identified and properly used. Electronic systems can be understood and explained with drawings/instructions		
.1	Explain the term 'high gain' in a control system					
.2	Explain how instability in a control system can occur					
.3	Sketch a diagrammatic arrangement of an automatic control system you have worked on showing the control elements					
.4	Give examples of Proportional-Integral-Derivative (PID) controllers that may be adjusted to achieve improved results/stability					
.5	List tuning methods commonly used on board					
.6	List software applications used in PID loop tuning					
.7	Explain the fundamental difference in control system for heating, ventilation and air conditioning systems					
.8	Give an example of a system where 'droop' has to be controlled					
.9	Describe the function of a PLC-based controller, identifying pre-set and adjustable parameters					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated	
6.	Competence: Operate electrical, electronic and control systems				Designated Training Officer/In Service Assessor (Initials/Date)	
6.9	Proportional-Integral-Derivative (PID) control characteristics			The instructions and manuals relevant for safe and efficient operations are quickly identified and properly used. Electronic systems can be understood and explained with drawings/instructions		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Explain the basic principle of three term control					
.2	Demonstrate a knowledge of PID control characteristics and associated system devices for process control					

Ref No	Training		Criteria for Evaluation		Competence Demonstrated	
7.	Competence: Maintenance and repair of electrical and electronic equipment				Designated Training Officer/In Service Assessor (Initials/Date)	
7.1	Locate and interpret electrical and simple electronic diagrams		Manuals and diagrams are quickly located and those selected are the most suitable for the task to be performed			
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)	Advice on Areas for Improvement		Task Completed Supervising Officer/ Instructor (Initials/Date)	
7.1	List shipboard equipment for which relevant manuals/ diagrams used: 1. 2. 3. 4. 5.					

7.2	Knowledge of construction and operation of electrical testing and measuring equipment			The selected measuring instruments and testing equipment are appropriate. Interpretation of results is checked for compliance with stated tolerances			
In the box below list the shipboard plant or equipment on which you have used the following: battery impedance testers, current clamp meters, multimeters, dielectric test sets, high/low voltage detectors and insulation testers.							
	Item Worked On	Measuring Instruments and Test Equipment Used	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1							
.2							
.3							
.4							
.5							
7.3	Ensure safety of all personnel working on electrical systems, including the safe isolation of electrical equipment, required before personnel are permitted to work on such equipment			Dismantling, inspecting, repairing and reassembling equipment are in accordance with manuals and good practice. Reassembling and performance testing is in accordance with manuals and good practice			
.1	Isolate and lock out electrical equipment, applying safety measures						
.2	Apply knowledge of safe use of electrical equipment for testing and maintenance in hazardous areas						
.3	Demonstrate an understanding of safe working practices and procedures including use of appropriate clothing for: Use of power operated tools						
.4	Entry into enclosed spaces (tank entry) with electrical equipment						
.5	Work on electrical switchboards						
.6	Use of lifting gear						
.7	Work within refrigeration machinery spaces						
.8	Work on electrical machinery						

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
7.	Competence: Maintenance and repair of electrical and electronic equipment					
7.3	Ensure safety of all personnel working on electrical systems, including the safe isolation of electrical equipment, required before personnel are permitted to work on such equipment (continued)			Dismantling, inspecting, repairing and reassembling equipment are in accordance with manuals and good practice. Reassembling and performance testing is in accordance with manuals and good practice		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.9	Explain the precautions to be taken when testing the insulation of generator cables and wiring connected to an automatic voltage regulator (AVR) unit					
.10	Explain why step down isolating transformers are sometimes used with portable tools and hand lamps					
7.4	Maintenance and repair of electrical system equipment, switchboards, electric motors, generator and DC electrical systems and equipment			Dismantling, inspecting, repairing and reassembling equipment are in accordance with manuals and good practice. Reassembling and performance testing is in accordance with manuals and good practice		
.1	Interpret accurately the information in a system diagram, a circuit diagram and a wiring diagram					
.2	Assist with routine checks and tests on electronic control systems					
.3	Demonstrate a knowledge of main switchboard and control room console layouts					
.4	Sketch a circuit diagram showing the arrangements for emergency battery charging for the ER alarm system					
.5	Demonstrate a knowledge of the vessel's emergency power requirements					
.6	Assist with main switchboard routine maintenance of contacts and connections					
.7	Demonstrate a knowledge of the procedure to split the main switchboard					
.8	Demonstrate a knowledge of switchboard instrumentation and safe working practices associated with its maintenance					

.9	Assist with measuring the insulation resistance of a generator					
.10	Explain why insulation testing is best conducted while hot, or at working temperature					
.11	Carry out insulation tests on a motor using a Megger					
.12	Assist in the maintenance of a starter					
.13	Sketch a circuit diagram showing the arrangements for battery charging					
.14	Carry out routine testing and maintenance on emergency storage batteries					
7.5	Detect and repair electrical faults and malfunctions and take measures to prevent damage			<i>Dismantling, inspecting, repairing and reassembling equipment are in accordance with manuals and good practice. Reassembling and performance testing is in accordance with manuals and good practice</i>		
.1	Sketch the circuit diagram for the earth indicator lamps on the main switchboard					
.2	Carry out Megger testing for insulation resistance and continuity testing					
.3	Assist with fault finding on electrical equipment control systems					
.4	Assist with tracing earth faults					
7.6	Repair faults and correct malfunctions			<i>Dismantling, inspecting, repairing and reassembling equipment are in accordance with manuals and good practice. Reassembling and performance testing is in accordance with manuals and good practice</i>		
.1	Assist with maintenance, repair and fault finding on electronic control systems. List items worked on: 1. 2. 3. 4. 5.					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
7.	Competence: Maintenance and repair of electrical and electronic equipment					
7.6	Repair faults and correct malfunctions (continued)			<i>Dismantling, inspecting, repairing and reassembling equipment are in accordance with manuals and good practice. Reassembling and performance testing is in accordance with manuals and good practice.</i>		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.2	Assist with maintenance, repair and fault finding on AC electrical systems. List items worked on: 1. 2. 3. 4. 5.					
.3	Assist with maintenance, repair and fault finding on DC electrical systems. List items worked on: 1. 2. 3. 4. 5.					
7.7	Detection of electric malfunction, location of faults and measures to prevent damage			<i>Dismantling, inspecting, repairing and reassembling equipment are in accordance with manuals and good practice. Reassembling and performance testing is in accordance with manuals and good practice.</i>		
.1	Demonstrate a knowledge of earth faults and how to avoid them					
.2	Assist in tracing and correcting earth faults					
.3	Isolate and lock out associated equipment when engaged in repair or maintenance work					

.4	Carry out routine testing and maintenance on alarm systems, ensuring that the circuits are isolated, locked out and protected by notices and that appropriate permit to work is issued					
.5	Assist with correct earthing-down routine for maintenance work on high voltage equipment					
.6	Assist with fault finding on ship's lighting circuits and component testing					
.7	Assist with repairing or replacing various types of accommodation lights, cargo hold and deck flood lights used on board					
7.8	Knowledge of the function and performance tests and configuration of monitoring systems, automatic control devices and protective devices			<i>Dismantling, inspecting, repairing and reassembling equipment are in accordance with manuals and good practice. Reassembling and performance testing is in accordance with manuals and good practice</i>		
.1	Explain why on any system there should be separate sensors for monitoring and control					
.2	Check and replace defective sensors essential for engine operation					
.3	State at least one main engine monitoring system that automatically stops the engine in case of a fault					
.4	Repair or replace: Fuses					
.5	Control lamps					
.6	Temperature sensors					
.7	Pressure sensors					
.8	Carry out routine testing and maintenance on: Circuit breakers					
.9	Tripping mechanisms					
.10	Motor starters					
.11	Lights					
.12	Check alarm settings and pre-sets contained in a system maintenance log					

Ref No	Training		Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
7.	Competence: Maintenance and repair of electrical and electronic equipment				
7.8	Knowledge of the function and performance tests and configuration of monitoring systems, automatic control devices and protective devices (continued)		Dismantling, inspecting, repairing and reassembling equipment are in accordance with manuals and good practice. Reassembling and performance testing is in accordance with manuals and good practice		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)
.13	Outline the advantages and disadvantages of DC and AC motors				
.14	Outline how an electronic drive control can stop a motor overloading but keep it operating				
.15	Explain where heat is generated in an electronic drive and how it is dissipated				

FUNCTION: MAINTENANCE AND REPAIR AT THE OPERATIONAL LEVEL

Ref No	Training		Criteria for Evaluation		Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
8.	Competence: Appropriate use of hand tools, machine tools and measuring instruments for fabrication and repair on board					
8.1	Knowledge of characteristics and limitations of materials and processes used in construction and repair of ships and equipment		The identification of important parameters for fabrication of typical ship-related components is appropriate. Selection of materials is appropriate. Fabrication is to designated tolerances			
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)	Advice on Areas for Improvement		Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Demonstrate a knowledge of characteristics, properties and limitations of: Mild steel					
.2	High tensile steel					
.3	Stainless steel					
.4	Brass					
.5	Aluminium alloy					
.6	Copper					
8.2	Knowledge of characteristics and limitations of processes used for fabrication and repair		The identification of important parameters for fabrication of typical ship-related components is appropriate. Selection of materials is appropriate. Fabrication is to designated tolerances			
.1	Demonstrate a knowledge of characteristics, properties and limitations of: Welding mild steel					
.2	Electric arc welding					
.3	Argon arc welding					
.4	Brazing					
.5	Riveting					
.6	Synthetic fixing methods					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)
8.	Competence: Appropriate use of hand tools, machine tools and measuring instruments for fabrication and repair on board				
8.3	Methods for carrying out safe emergency/temporary repairs			Use of equipment and hand tools, machine tools and measuring instruments is appropriate and safe	
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)
.1	Demonstrate temporary repair to leaking pipe				
.2	List other temporary repairs made 1. 2. 3. 4. 5.				
8.4	Safety measures to be taken to ensure a safe working environment and for using hand tools, machine tools and measuring instruments			Use of equipment and hand tools, machine tools and measuring instruments is appropriate and safe	
.1	Demonstrate an understanding of safe working practices and procedures for: Use of power operated tools				
.2	Machine tools				
.3	Welding equipment				
.4	Don appropriate personal protective equipment				
8.5	Use of hand tools and machine tools			Use of equipment and hand tools, machine tools and measuring instruments is appropriate and safe	
In the box below list the shipboard plant or equipment on which you have used, for example, the following hand tools: chisels, saws, spanners, files, hand-drills and machine tools, drill press, milling machine, saw, grinding machine and abrasive wheel.					
	Item Fabricated or Repaired	Hand Tools/Machine Tools Used	Task Completed Supervising Officer/ Instructor (Initials/Date)	Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)
.1					

.2							
.3							
.4							
.5							
.6							
.7							
.8							
.9							
.10							
8.6	Use of measuring instruments				The selected measuring instruments used for repair and maintenance of machinery and equipment are relevant for the tasks; correct measurements are taken and checked for compliance with stated tolerances		

In the box below list the shipboard plant or equipment on which you have used measuring equipment. Measuring instruments may include, for example, oddleg calipers, digital calipers, internal micrometer, depth gauge and vernier gauge.

	Item Fabricated or Repaired	Measuring Instruments Used	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1							
.2							
.3							
.4							
.5							
.6							

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
8.	Competence: Appropriate use of hand tools, machine tools and measuring instruments for fabrication and repair on board					
8.6	Use of measuring instruments (continued)			The selected measuring instruments used for repair and maintenance of machinery and equipment are relevant for the tasks; correct measurements are taken and checked for compliance with stated tolerances		
	Item Fabricated or Repaired	Measuring Instruments Used	Task Completed Supervising Officer/ Instructor (Initials/Date)	Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.7						
.8						
.9						
.10						
8.7	Use of sealants and packings			The selection of materials is appropriate		
.1	Demonstrate the correct use of various types of sealants and packings and note when used: Gland seals					
.2	Flanges					
.3	Gasketing materials, including compressed non-asbestos, cork, rubber and fibre					
.4	Valve stem packing					
.5	Pump seal					
.6	Hydraulic seal					
.7	O-Ring seal					
.8	Flange joint sealants					
.9	Exhausts and piping					
.10	Shaft seals/packing					

8.8	Use of special tools for fabrication and repair work on board	Correct tools are chosen and used in accordance with instructions, manuals and safe working practice		
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In the box below list the machinery or equipment on which you have carried out repairs, or assisted in repairs with others, showing the special tools used. This includes dismantling, inspection, repair and reassembly work. As a minimum carry out repairs requiring use of: hydraulic tools, bearing pullers and torque wrench. Machinery may include steering gear, engine room pumps and fans, deck winches and windlass, galley and catering equipment and air conditioning.

	Item Fabricated or Repaired	Special Tools Used	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1							
.2							
.3							
.4							
.5							
.6							
.7							
.8							
.9							
.10							

8.9	Use of machine tools and welding equipment for fabrication and repairs	The selected material is suitable for the part to be fabricated and the work is carried out within the designated tolerances and in accordance with safe working practice		
-----	--	---	--	--

In the box below list the parts which you have fabricated (made), or assisted in making, using machine tools. For example: centre lathes, drill press, gas welding/brazing equipment, gas cutting equipment including plasma arc, electric arc welding and other special equipment.

	Item Fabricated or Repaired	Machine Tools or Equipment Used	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1							
.2							

Ref No	Training	Criteria for Evaluation	Competence Demonstrated	
8.	Competence: Appropriate use of hand tools, machine tools and measuring instruments for fabrication and repair on board		Designated Training Officer/In Service Assessor (Initials/Date)	
8.9	Use of machine tools and welding equipment for fabrication and repairs (continued)	<i>The selected material is suitable for the part to be fabricated and the work is carried out within the designated tolerances and in accordance with safe working practice</i>		

In the box below list the parts which you have fabricated (made), or assisted in making, using machine tools. For example: centre lathes, drill press, gas welding/brazing equipment, gas cutting equipment including plasma arc, electric arc welding and other special equipment.

	Item Fabricated or Repaired	Measuring Instruments Used	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.3							
.4							
.5							
.6							
.7							
.8							
.9							
.10							

Ref No	Training	Criteria for Evaluation		Competence Demonstrated	
9.	Competence: Maintenance and repair of shipboard machinery and equipment			Designated Training Officer/In Service Assessor (Initials/Date)	
9.1	Locate and use relevant data sources, manuals and drawings Before starting any maintenance or repair work ensure that you have completed the tasks concerned with Safety at Work on page 26. In addition, ensure that you are familiar with the procedures for safe isolation of electrical equipment for your present ship, and that you are in possession of an appropriate permit to work.	<i>The manufacturers' instructions and drawings relevant for the job are quickly identified and properly used</i>			
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	
.1	Demonstrate a knowledge of the ship's Planned Maintenance System				
.2	Demonstrate an understanding of manufacturers' instructions and drawings for use in maintenance tasks				
.3	State what is required in a Planned Maintenance System				
.4	Assist with input to the ship's Planned Maintenance System				
.5	Retrieve reports from a computer-based maintenance system				
.6	Participate in a survey of running machinery using condition monitoring equipment, if applicable				
.7	Assist in interpretation of results of such survey				
.8	Describe how items of spare gear are stored and maintained in good condition				
9.2	Ensure safety of all personnel working on plant or equipment	<i>Isolation, dismantling and reassembly of plant and equipment is in accordance with accepted safe working practices and procedures</i>			
.1	State special precautions to be taken for repair and maintenance work in hazardous areas				
.2	Demonstrate an understanding of safe working practices and procedures for: Use of portable power operated tools				

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
9.	Competence: Maintenance and repair of shipboard machinery and equipment					
9.2	Ensure safety of all personnel working on plant or equipment (continued)			Isolation, dismantling and reassembly of plant and equipment is in accordance with accepted safe working practices and procedures		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.3	Entry into enclosed spaces (tank entry)					
.4	Work beneath floor plates					
.5	Use of lifting gear					
.6	Moving heavy machinery					
.7	Work within refrigeration machinery spaces					
.8	Work on electrical machinery					
.9	Disposal of oily waste materials					
.10	Use of appropriate protective clothing					
.11	Working at height					
.12	Manual lifting and carrying					
9.3	Undertake maintenance and repairs to the main engine			Dismantling, inspecting, repairing and reassembling equipment is in accordance with manuals and good practice. Recommissioning and performance testing is in accordance with manuals and good practice. Selection of materials and parts is appropriate		
.1	Take and log readings of crankshaft deflections					
.2	Change, inspect, check condition, wear and clearance, overhaul and test, as appropriate: Inlet valves					
.3	Fuel injection valves					

.4	Air start valves				
.5	Relief valves				
.6	Exhaust valves, where appropriate				
.7	Fuel pumps				
.8	Cam shafts				
.9	Crosshead bearings				
.10	Fuel oil filters				
.11	Lube oil filters				
.12	Air filters				
.13	Use turning gear, under supervision, taking all safety precautions				
.14	Change and/or overhaul the following main engine components, checking clearances, where appropriate: Pistons				
.15	Cylinder heads				
.16	Turbochargers				
.17	Top end bearings				
.18	Bottom end bearings				
.19	Indicator cocks				
.20	Main bearings				
.21	Piston-rod scraper box/stuffing box				
.22	Crosshead guides				
.23	Tie bolts				
.24	Holding-down bolts and chocks				

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
9.	Competence: Maintenance and repair of shipboard machinery and equipment					
9.3	Undertake maintenance and repairs to the main engine (continued)			<i>Dismantling, inspecting, repairing and reassembling equipment is in accordance with manuals and good practice. Recommissioning and performance testing is in accordance with manuals and good practice. Selection of materials and parts is appropriate</i>		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.25	Inspect scavenge trunk and exhaust spaces and report on: Cleanliness/deposits					
.26	Scavenge drains					
.27	Scavenge valves					
.28	Carry out a crankcase inspection					
9.4	Undertake maintenance and repair to the auxiliary engine			<i>Dismantling, inspecting, repairing and reassembling equipment is in accordance with manuals and good practice. Recommissioning and performance testing is in accordance with manuals and good practice. Selection of materials and parts is appropriate</i>		
.1	Take and log readings of crankshaft deflections					
.2	Change, inspect, check condition, wear and clearance, overhaul and test: Fuel injection valves					
.3	Air start valves					
.4	Relief valves					
.5	Inlet valves					
.6	Exhaust valves					
.7	Fuel pumps					
.8	Camshaft					

.9	Fuel oil filters					
.10	Lube oil filters					
.11	Air filters					
.12	Jacket cooling water pump					
.13	Change and/or overhaul the following components, checking and adjusting clearances, where appropriate: Pistons					
.14	Cylinder heads					
.15	Turbochargers					
.16	Top end bearings					
.17	Bottom end bearings					
.18	Indicator cocks					
.19	Main bearings					
.20	Holding-down bolts and chocks					
.21	Carry out a crankcase inspection					
.22	Commission engine after overhaul					
9.5	Undertake maintenance and repair to the auxiliary boiler			<i>Dismantling, inspecting, repairing and reassembling equipment is in accordance with manuals and good practice. Recommissioning and performance testing is in accordance with manuals and good practice. Selection of materials and parts is appropriate</i>		
.1	Take a boiler out of service					
.2	Isolate boiler					
.3	Blow a boiler down					
.4	Open up a boiler					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
9.	Competence: Maintenance and repair of shipboard machinery and equipment					
9.5	Undertake maintenance and repair to the auxiliary boiler (continued)			<i>Dismantling, inspecting, repairing and reassembling equipment is in accordance with manuals and good practice. Recommissioning and performance testing is in accordance with manuals and good practice. Selection of materials and parts is appropriate.</i>		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.5	Examine a boiler, reporting on its condition: Internally					
.6	Externally					
.7	Open up and inspect: Safety valves					
.8	Feed check valves					
.9	Ancillary valves					
.10	Overhaul and test water gauge glass and check that passages, cocks and valves are clear					
.11	Change and overhaul burner					
9.6	Undertake maintenance and repair to plant and equipment			<i>Dismantling, inspecting, repairing and reassembling equipment is in accordance with manuals and good practice. Recommissioning and performance testing is in accordance with manuals and good practice. Selection of materials and parts is appropriate.</i>		
.1	Open up purifiers/separators for cleaning and maintenance					
.2	Reassemble purifiers/separators					
.3	Carry out routine maintenance on a main compressor					
.4	Check and service: Control air filters					
.5	Control air driers, replacing desiccant					

.6	Carry out routine maintenance on refrigeration plant					
.7	Carry out routine maintenance on fresh water generator					
.8	Open up and overhaul positive displacement pump					
.9	Open up and overhaul centrifugal pump					
.10	Overhaul and test valves including: Gate					
.11	Stop disk non return					
.12	Screw lift					
.13	Relief					
.14	Two or three way					
.15	Shut-off cock					
.16	Carry out routine maintenance on: Anchor windlass					
.17	Cargo winches					
.18	Cargo cranes					
.19	Mooring winches					
.20	Capstans					
.21	Hatch covers					
.22	Steering gear					
.23	Engine room lifting gear					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
9.	Competence: Maintenance and repair of shipboard machinery and equipment					
9.6	Undertake maintenance and repair to plant and equipment (continued)			<i>Dismantling, inspecting, repairing and reassembling equipment is in accordance with manuals and good practice. Recommissioning and performance testing is in accordance with manuals and good practice. Selection of materials and parts is appropriate</i>		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.24	List other items of plant and equipment on which you have worked: 1. 2. 3. 4. 5.					
9.7	Undertake maintenance and repair to emergency equipment			<i>Isolation, dismantling and reassembly is in accordance with accepted practices and procedures. Correct tools are chosen and used without causing damage to machinery or equipment</i>		
.1	Carry out routine maintenance on: Fire pumps					
.2	Fire flaps					
.3	ER fire extinguishing system and equipment					
.4	Emergency generator					
.5	Emergency compressor					
.6	Remote stops for pumps with overboard discharges					
.7	Fuel valve trips					
.8	Breathing apparatus sets and recharging breathing apparatus bottles					
.9	Survival craft					

FUNCTION: CONTROLLING THE OPERATION OF THE SHIP AND CARE FOR PERSONS ON BOARD AT THE OPERATIONAL LEVEL

Ref No	Training			Criteria for Evaluation	Competence Demonstrated	
10.	Competence: Application of leadership and teamworking skills				Designated Training Officer/In Service Assessor (Initials/Date)	
10.1	Plays team role			Displays awareness of others working nearby and in common goals. Communicates clearly and unambiguously in language understood. Challenges questionable decisions in a seamanlike manner. Freely shares information concerning the manoeuvre or task in hand		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Understand that as a team member everyone has different experience and has a role to play in any task					
.2	Participate actively in task planning meetings involving different ranks					
.3	Understand communication is a two-way exchange and demonstrate this in practice both in the engine room and when working on deck					
.4	Maintain awareness of changing situations					
.5	Accept authority but not be afraid to question if in doubt					
.6	Check own understanding of situation is shared by other team members					
.7	Participate actively in task review and evaluation meetings involving different ranks					
10.2	Exhibits leadership ability			Takes initiative and carries others along with what needs to be done in timely way		
.1	Think ahead and plan tasks that will follow the immediate task or manoeuvre					
.2	Set priorities correctly when seeing conflict between immediate needs and tasks that may be held back					
.3	Allocate resources effectively to achieve desired outcomes					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
10.	Competence: Application of leadership and teamworking skills					
10.2	Exhibits leadership ability (continued)			Takes initiative and carries others along with what needs to be done in timely way		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.4	Check results and take corrective actions as needed/ instructed					
.5	Demonstrate confidence and maturity to refer to senior officer if in doubt					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
11.	Competence: Ensure compliance with pollution prevention requirements					
11.1	Implement proactive measures to protect the marine environment			<i>The operations are properly planned and comply with international regulation in spirit as well as in word. Ensures that a positive environmental reputation is maintained</i>		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Understand that environmental protection includes both sea and air which are protected by mandatory MARPOL regulations					
.2	Name at least two Particularly Sensitive Sea Areas (PSSAs) 1. 2.					
.3	Demonstrate by example preparedness to take personal responsibility for actions to protect the marine environment					

.4	Understand that marine pollutants must be landed ashore for safe disposal in compliance with MARPOL					
.5	Understand there are strict rules covering the storage and disposal of oily water mixtures applicable to all ships					
.6	Understand there are strict rules covering disposal of noxious liquid substances applicable to all ships					
.7	Understand there are strict rules covering disposal of harmful substances carried in packaged form applicable to all ships					
.8	Understand there are strict rules covering pollution prevention by sewage applicable to ships					
.9	Understand there are strict rules for prevention of pollution by garbage from ships, applicable to all ships					
.10	Understand there are strict rules covering air pollution from ships at sea which will progressively apply to all ships					
.11	Understand the impact of SOx, NOx, VOC and PM and why efforts are needed to reduce atmospheric pollution					
11.2	Ensure that procedures are agreed and properly planned before bunkering			<i>The operations are properly planned, all scuppers are blocked and pipes and hoses inspected before bunkering takes place</i>		
.1	Plug deck scuppers					
.2	Demonstrate knowledge of ship's bunkering procedures					
.3	Participate in bunkering operations					
.4	Demonstrate the emergency shutdown procedure					
11.3	Initiate immediate investigation to detect the source on discovering any pollution around the ship			<i>All available resources are utilised to detect the source and the master or authorities are informed as appropriate</i>		
.1	Participate in an emergency response exercise for controlling spillage of oil or other noxious or toxic substances on board					
.2	Be aware of the importance of immediately reporting and investigating potential pollution incidents					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated	
11.	Competence: Ensure compliance with pollution prevention requirements				Designated Training Officer/In Service Assessor (Initials/Date)	
11.4	Stop or prevent leakages and spills of harmful liquids and solid substances			The situation is thoroughly assessed and the actions taken are well organised and exercised and due consideration taken of the extent of the pollution		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Demonstrate use of Material Safety Data Sheets and the IMDG Code to obtain information on cargo hazards and handling instructions					
.2	Participate in drill for clean-up of hazardous spillage					
11.5	Sound all tanks and compartments if any damage is suspected			The soundings are readily available and the results immediately reported to the master		
.1	Participate in an emergency response exercise for stranding					
.2	Perform soundings of bilges, peak tanks, double bottom and other tanks and record information if any hull damage is suspected					
11.6	Carry out bilge, ballast and bunkering operations			All operations are carried out in accordance with MARPOL and due regard paid to the Shipboard Oil Pollution Emergency Plan (SOPEP)		
.1	Locate the ship's ballast water management plan and demonstrate an understanding of its content					
.2	Understudy the engineer officer conducting a ballasting operation					
.3	Have knowledge of requirements of MARPOL and Annexes					
.4	Demonstrate an understanding of Oil Discharge Monitor Equipment operation (oil tankers)					
.5	Demonstrate an understanding of the record keeping required in the Oil Record Book					

Ref No	Training		Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
12.	Competence: Maintain seaworthiness of the ship				
12.1	Inspect hull and hull openings, compartments, hatch covers and equipment, and take action where defects are detected		The inspection is properly carried out, due regard paid to the prevailing circumstances and areas where defects are most likely to occur. Any defect is immediately reported and recorded and the suggested or executed action adequate for the situation		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)	Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Demonstrate an understanding of the precautions required for: Entry into enclosed spaces				
.2	Working at height				
.3	Using power tools				
.4	Manual lifting and carrying				
.5	Where applicable, assist with the opening, closing and securing of hatches				
.6	Assist with the maintenance of watertight doors, ports and hatches				
.7	Carry out routine maintenance and repair on: Anchor windlass				
.8	Cargo handling equipment				
.9	Mooring winches				
.10	Carry out full inventory check of the engine stores				
.11	Prepare steel plates and other surfaces for protective coating				
.12	Apply protective coats to appropriate surfaces				

Ref No	Training			Criteria for Evaluation	Competence Demonstrated	
12.	Competence: Maintain seaworthiness of the ship				Designated Training Officer/In Service Assessor (Initials/Date)	
12.2	Ensure that all loose objects are securely fastened to avoid damage					
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Ensure that all gear, tools, spares etc. are properly stowed and secured					
12.3	Arrange for regular control measures to ensure watertight integrity			Peaks, bilges, tanks and other compartments are sounded regularly, the results recorded and any irregularities reported and examined further		
.1	Take and record the daily soundings of engine room tanks, bilges and other spaces: By manual means					
.2	By use of gauges					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated	
13.	Competence: Prevent, control and fight fires on board				Designated Training Officer/In Service Assessor (Initials/Date)	
13.1	Operate fire and smoke detecting equipment					
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Understand use and assist in the maintenance of: Portable foam extinguisher					
.2	Portable CO ₂ extinguisher					
.3	Portable dry powder extinguisher					

.4	Portable water extinguisher					
.5	Maintain hoses, nozzles and couplings					
13.2	Ensure that all persons on watch are able to detect and correct hazardous situations and actions and keep the ship clean and tidy			<i>Personnel on watch making inspections in areas at risk from possible fires are supervised. Ensure readily combustible materials are stored safely and the watch demonstrate an attitude of alertness to fire prevention</i>		
.1	Perform fire patrol duties					
.2	Re-stow gear and secure after maintenance work					
13.3	Instruct the watch in locating fire-fighting appliances and emergency escape routes and sound alarm			<i>Instruct watch in use of portable or other fire extinguishers. Demonstrate an ability to raise the alarm</i>		
.1	Carry out a full inspection of fire-fighting equipment and report to the chief engineer					
.2	Participate in an emergency response exercise for fire at sea and in port					
13.4	Locate fire stations and demonstrate proper use of fixed installations and other fire-fighting appliances and agents			<i>All stations are located and the most suitable one selected in the event of a fire. Proper equipment and extinguishing agents for the various materials on fire are selected</i>		
.1	Assist with the testing of the following systems, where fitted: Fire detection and alarm systems					
.2	Fire alarms					
.3	Fixed automatic sprinklers					
.4	Fixed steam systems					
.5	Fixed foam extinguishers					
.6	Fixed CO ₂ systems					
.7	Fire flaps and dampers					
.8	Automatic and manual fire doors					
.9	Emergency shut off valves, pump stops and main engine stops					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
13.	Competence: Prevent, control and fight fires on board					
13.4	Locate fire stations and demonstrate proper use of fixed installations and other fire-fighting appliances and agents (continued)			All stations are located and the most suitable one selected in the event of a fire. Proper equipment and extinguishing agents for the various materials on fire are selected		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.10	Describe the operation of the fixed fire extinguishing system for the engine room					
.11	State the safety precautions required prior to operating the system					
13.5	Locate and use fire protective equipment (fire-fighter's outfit, including breathing apparatus)			The equipment is quickly donned and used in such a way that no accidents are likely to occur		
.1	Demonstrate the procedures and precautions required for entry into an enclosed space					
.2	Recognise the different uses for a Self Contained Breathing Apparatus (SCBA) set and an Emergency Escape Breathing Device					
.3	Demonstrate donning and use of SCBA sets					
.4	Demonstrate donning and use of a fire-fighter's outfit					
.5	Demonstrate donning and use of a fire-fighter's outfit with a SCBA set					
.6	Demonstrate the use of a SCBA record/control board					
13.6	Demonstrate ability to act in accordance with the fire-fighting plan during fire drills			During debriefing after an exercise or a real fire extinguishing incident, the reasons for each action taken, including the priority in which they were taken, are explained and accepted as the most appropriate		
.1	Take charge of a fire party during an exercise					
.2	Demonstrate the use and location of all engine room safety appliances					
.3	Demonstrate a knowledge of all engine room escape routes					

Ref No	Training	Criteria for Evaluation		Competence Demonstrated	
14.	Competence: Operate life-saving appliances			Designated Training Officer/In Service Assessor (Initials/Date)	
14.1	Organise abandon ship drills	On sounding the alarm all persons meet at the designated lifeboat station wearing safety belts or immersion suits and carry out their duties on request			
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	
.1	Understand the hazards to seafarers of manning life boats for drills and exercises				
.2	Understand the need to be familiar with the operation of on-load release mechanisms				
.3	Recognise that fall prevention devices (FPDs), where fitted, should be used in drills (to prevent unforeseen detachment)				
.4	Recognise the need for meticulous inspection and maintenance of on-load release mechanisms				
.5	Identify the permanent marking on survival craft with regard to the number of occupants				
.6	Locate and test the operation of: Radio devices including satellite EPIRBs and SARTs				
.7	Pyrotechnic distress signals				
.8	State precautions for disposal of out of date pyrotechnics				
.9	Understudy an officer in charge of an abandon ship drill				
14.2	Launch, handle and recover a lifeboat	Correct orders for embarkation, launching and immediately clearing the ship's side are given. The boat is safely handled under motor or oars, as appropriate. The boat is safely recovered and ready			
.1	Assist with preparation and swinging out of lifeboats and be aware of potential risks				
.2	Assist with preparation and boarding of free fall lifeboat and be aware of potential risks				

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
14.	Competence: Operate life-saving appliances					
14.2	Launch, handle and recover a lifeboat (continued)			Correct orders for embarkation, launching and immediately clearing the ship's side are given. The boat is safely handled under motor or oars, as appropriate. The boat is safely recovered and ready		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.3	Assist with lowering a lifeboat to clear the ship and ride to a sea anchor					
.4	Start and operate a lifeboat engine					
.5	Crew a boat under: Oars ☐ Power ☐					
.6	Cox a boat under: Oars ☐ Power ☐					
.7	Assist with recovering and securing a lifeboat					
.8	Assist with recovering and securing a free fall lifeboat					
14.3	Launch or throw overboard a life raft, and manoeuvre it clear of ship's side			The duties for the person designated for the raft are clearly allocated, orders efficiently executed, the raft is quickly righted if inverted, and all persons boarded before the raft moves away from the ship		
.1	Demonstrate an understanding of the procedure for launching and inflating life rafts, if the opportunity arises					
14.4	Operate radio life-saving appliances			Radio contact is established without alerting anybody by transmitting false signals		
.1	Rig and operate the portable lifeboat radio under supervision					

14.5	Ensure that all required equipment on board a rescue craft is functioning and maintained as specified in the SOLAS Training Manual			Proper use of pyrotechnics, food, water and signalling equipment is satisfactorily demonstrated		
.1	Demonstrate an understanding of statutory equipment required in survival craft and its correct use					
.2	State minimum food and water requirements for occupants of survival craft					
.3	Locate and understand operation of pyrotechnics including precautions for their disposal					
.4	Explain the operation of rocket line throwing apparatus					
.5	Explain the operation of distress rockets, flares and other pyrotechnics					
.6	Assist with the maintenance of: Lifeboats and rescue boats					
.7	Lifeboat equipment and provisions					
.8	Launching davits and gear					
.9	Buoyant apparatus, e.g. lifebuoys, lifejackets and attachments					
.10	Immersion suits and thermal protective aids					
.11	Other survival craft, specify type					
.12	Assist with the routine maintenance of a lifeboat engine					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
15.	Competence: Apply medical first aid on board ship					
15.1	Stop excessive bleeding, ensure breathing and put casualties in proper position			<i>The actions demonstrated are in compliance with accepted recommendations given in international medical first aid guidance</i>		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Participate in an emergency first aid drill at sea					
.2	Demonstrate a basic understanding of first aid principles: Stopping bleeding					
.3	Treatment of suffocation/drowning					
.4	Placing casualty in the recovery position					
15.2	Detect signs of shock and heat stroke and act accordingly			<i>The treatment recommended or given is adequate. Ability to request Radio Medico for advice is demonstrated</i>		
.1	Demonstrate how to handle a casualty in shock					
.2	Demonstrate procedures for dealing with heat stroke					
15.3	Treat burns, scalds, fractures, and hypothermia			<i>Recommended guidelines for proper actions are explained. Principles for avoiding hypothermia are demonstrated</i>		
.1	State procedure for dealing with a casualty of electric shock					
.2	Demonstrate procedure for treating burns					
.3	Demonstrate procedure for treating minor fractures					
.4	State procedures for avoiding hypothermia					
.5	Demonstrate procedures for treating casualty with hypothermia					

Ref No	Training			Criteria for Evaluation	Competence Demonstrated Designated Training Officer/In Service Assessor (Initials/Date)	
16.	Competence: Monitor compliance with legislative requirements					
16.1	State where laws, rules and regulations concerning ship operation and pollution prevention are available			The statement given is correct and includes relevant bodies or organisations which may be contacted to obtain special information or guidance which is not easily accessible		
	Task/Duty	Task Completed Supervising Officer/ Instructor (Initials/Date)		Advice on Areas for Improvement	Task Completed Supervising Officer/ Instructor (Initials/Date)	
.1	Locate on board copies of: SOLAS					
.2	MARPOL					
.3	Shipboard Oil Pollution Emergency Plan (SOPEP)					
.4	Garbage Record Book					
.5	Locate copies of certificates issued under SOLAS, MARPOL, Load Line, STCW , ILO MLC, and other regulations					
16.2	Use legislation to check on board operations comply with international regulations			Correct response is established within an acceptable period of time and consequential actions executed		
.1	Participate in engine room oil and oily waste handling operations in compliance with MARPOL					
.2	Dispose of garbage in compliance with MARPOL and ship's Garbage Management Plan					
.3	Assist in checking machinery and equipment prior to survey					
.4	Participate in shipboard inspection prior to an International Oil Pollution Prevention (IOPP) survey					
16.3	Searching for stowaways			A comprehensive and thorough search is conducted and findings reported to the responsible officer		
.1	Carry out a stowaway search					

SECTION 8 PROJECT WORK

INTRODUCTION

The object in carrying out projects during sea service is to ensure that you progressively gain a knowledge of the ships in which you serve and the equipment and life saving appliances carried on board.

To achieve this objective successfully will require intelligent observation, initiative and reference, where appropriate, to ships' plans and other data, besides manufacturers' instructions and operation manuals. It will, moreover, be essential in a number of cases to seek the help and guidance of your officers in order to obtain the information required.

The majority of the projects deal with obtaining factual details which concern such matters as the ship's structural features and equipment including various supply systems - bunker fuel, fresh and salt water, etc.

The technical accuracy of each project will be checked by the chief engineer and will be evaluated by your company and/or by your nautical college. The evaluation will look for:

- (a) Accuracy of information in written text, illustrations or calculations
- (b) Coverage of subject showing depth of research and logical presentation of facts
- (c) Neatness of writing, diagrams/labels
- (d) Spelling and grammar

PROJECTS

1. Pipe Systems Trace and make line diagrams of the following pipe systems. Use the correct symbols to show on the appropriate diagrams: valves (NRV, SDNR, etc.), remote or emergency controls and other arrangements.

Identify pressure relief valves, bursting discs, drains, air cocks, filter units, sounding arrangements and vent pipes.

- | | |
|--|--|
| 1. Main sea water | 9. Auxiliary steam |
| 2. Bilge, including OWS and emergency bilge pumping arrangements | 10. Main engine fuel oil system |
| 3. Fire main | 11. Auxiliary engine fuel oil system |
| 4. Ballast | 12. Main engine lube oil service |
| 5. Domestic fresh water (high temperature, low temperature) | 13. Sewage system |
| 6. Fuel-transfer system, (HFO and MDO), including remote or emergency controls and overflow arrangements for fuel transfer | 14. Compressed air systems for engine room and deck services |
| 7. Main steam | 15. Drain valves |
| 8. Feed-water | 16. Air cocks |
| | 17. Domestic refrigeration system |

2. Scale Drawings Draw approximately to scale:

- (a) A longitudinal section through the centre line of your ship showing and naming cargo holds (tanks), bunker, ballast and all other compartments/spaces,
- (b) A plan of each of the decks showing and naming accommodation, store rooms etc.

3. Safety On the plan of machinery spaces drawn above:

- (a) Show the position by key letters of each type of life-saving and fire-fighting equipment,
- (b) List the above key letters used in (a) and alongside each one give a brief description of each item.

4. Protection of the Marine Environment Summarise the company's policy on environmental protection. What measures are taken aboard your ship to minimise the risk of pollution. This includes the disposal of plastics, galley waste, noise, smoke, oil, sludge, sewage, grey water etc. Investigate and list the MARPOL regulations that aim to control and protect the marine environment.

5. Main Engine Make a line diagram of the lubricating system for the main engine. Indicate the types of valves, pumps and filters fitted. Show, with the aid of a diagram, the general lube oil distribution.

What is the average lube oil consumption of cylinder and crankcase oil? Why does this loss occur?

6. Steering Gear Describe the emergency operation of the steering gear.

7. Electrical Systems Describe the procedure for paralleling the ship's alternators or generators. Explain how load sharing is achieved.

8. Bunkering Describe the procedures for taking bunkers. State clearly the sequence of events and the precautions taken. Evaluate the results of any tests taken at the time or from analysis made by a laboratory ashore.

9. Bridge Watches Describe very briefly the purpose and functions of the main items of bridge equipment. Observe procedures and assist on the bridge during manoeuvring operations

- (a) Entering port
- (b) Leaving port
- (c) When anchoring or weighing anchor, and
- (d) During one watch at sea.

Describe how orders are given, confirmed and executed and the interactions with the engine room and other parts of the vessel.

SECTION 9 TASK SUMMARY CHART

The purpose of the summary chart is to provide you, your company and your ships' officers with a guide and continuous check on the numbers of tasks or duties listed in Section 7 that you have completed, and those that remain outstanding. Tick off only those tasks which you have completed. In the charts below, the green boxes simply indicate the start of a new group of tasks or duties.

FUNCTION - Marine Engineering at the Operational Level

1. COMPETENCE - Maintain a safe engineering watch

1.1.1	1.1.2	1.1.3	1.1.4	1.2.1	1.2.2	1.2.3	1.2.4	1.2.5	1.2.6	1.2.7	1.2.8	1.2.9	1.2.10	1.2.11	1.2.12
1.2.13	1.2.14	1.2.15	1.2.16	1.2.17	1.2.18	1.3.1	1.3.2	1.3.3	1.3.4	1.3.5	1.3.6	1.3.7	1.3.8	1.3.9	1.3.10
1.4.1	1.4.2	1.4.3	1.4.4	1.4.5	1.5.1	1.5.2	1.5.3	1.5.4	1.5.5	1.5.6	1.6.1	1.6.2	1.6.3	1.6.4	1.6.5
1.6.6															

2. COMPETENCE - Use English in written and oral form

2.1.1	2.1.2	2.2.1	2.2.2	2.2.3	2.2.4	2.2.5

3. COMPETENCE - Use internal communication systems

3.1.1	3.1.2	3.1.3	3.1.4	3.1.5	3.1.6

4. COMPETENCE - Operate main and auxiliary machinery and associated control systems

4.1.1	4.1.2	4.1.3	4.1.4	4.1.5	4.1.6	4.1.7	4.2.1	4.2.2	4.2.3	4.2.4	4.2.5	4.2.6	4.2.7	4.2.8	4.2.9
4.2.10	4.2.11	4.2.12	4.2.13	4.2.14	4.2.15	4.2.16	4.2.17	4.2.18	4.2.19	4.2.20	4.2.21	4.2.22	4.2.23	4.2.24	4.2.25
4.2.26	4.2.27	4.2.28	4.2.29	4.2.30	4.2.31	4.2.32	4.2.33	4.2.34	4.2.35	4.2.36	4.2.37	4.2.38	4.2.39	4.2.40	4.2.41
4.2.42	4.2.43	4.2.44													

5. COMPETENCE - Operate fuel, lubrication, ballast and other pumping systems and associated control systems

5.1.1	5.1.2	5.1.3	5.1.4	5.1.5	5.1.6	5.1.7	5.1.8	5.1.9	5.1.10	5.1.11	5.1.12	5.1.13	5.2.1	5.2.2	5.2.3
5.2.4	5.2.5	5.2.6	5.2.7	5.2.8	5.2.9	5.2.10	5.2.11	5.2.12	5.2.13	5.2.14	5.2.15	5.2.16	5.2.17		

FUNCTION - Electrical, Electronic and Control Engineering at the Operational Level

6. COMPETENCE - Operate electrical, electronic and control systems

6.1.1	6.1.2	6.1.3	6.1.4	6.1.5	6.1.6	6.2.1	6.2.2	6.2.3	6.2.4	6.2.5	6.2.6	6.2.7	6.2.8	6.2.9	6.3.1
6.3.2	6.3.3	6.3.4	6.4.1	6.4.2	6.4.3	6.4.4	6.5.1	6.5.2	6.5.3	6.5.4	6.6.1	6.6.2	6.6.3	6.6.4	6.6.5
6.7.1	6.7.2	6.7.3	6.7.4	6.8.1	6.8.2	6.8.3	6.8.4	6.8.5	6.8.6	6.8.7	6.8.8	6.8.9	6.9.1	6.9.2	

7. COMPETENCE - Maintenance and repair of electrical and electronic equipment

[illegible]

FUNCTION - Maintenance and Repair at the Operational Level

8. COMPETENCE - Appropriate use of hand tools, machine tools and measuring instruments for fabrication and repair on board

8.1.1	8.1.2	8.1.3	8.1.4	8.1.5	8.1.6	8.2.1	8.2.2	8.2.3	8.2.4	8.2.5	8.2.6	8.3.1	8.3.2	8.4.1	8.4.2
8.4.3	8.4.4	8.5.1	8.5.2	8.5.3	8.5.4	8.5.5	8.5.6	8.5.7	8.5.8	8.5.9	8.5.10	8.6.1	8.6.2	8.6.3	8.6.4
8.6.5	8.6.6	8.6.7	8.6.8	8.6.9	8.6.10	8.7.1	8.7.2	8.7.3	8.7.4	8.7.5	8.7.6	8.7.7	8.7.8	8.7.9	8.7.10
8.8.1	8.8.2	8.8.3	8.8.4	8.8.5	8.8.6	8.8.7	8.8.8	8.8.9	8.8.10	8.9.1	8.9.2	8.9.3	8.9.4	8.9.5	8.9.6
8.9.7	8.9.8	8.9.9	8.9.10												



9. COMPETENCE - Maintenance and repair of shipboard machinery and equipment

9.1.1	9.1.2	9.1.3	9.1.4	9.1.5	9.1.6	9.1.7	9.1.8	9.2.1	9.2.2	9.2.3	9.2.4	9.2.5	9.2.6	9.2.7	9.2.8
9.2.9	9.2.10	9.2.11	9.2.12	9.3.1	9.3.2	9.3.3	9.3.4	9.3.5	9.3.6	9.3.7	9.3.8	9.3.9	9.3.10	9.3.11	9.3.12
9.3.13	9.3.14	9.3.15	9.3.16	9.3.17	9.3.18	9.3.19	9.3.20	9.3.21	9.3.22	9.3.23	9.3.24	9.3.25	9.3.26	9.3.27	9.3.28
9.4.1	9.4.2	9.4.3	9.4.4	9.4.5	9.4.6	9.4.7	9.4.8	9.4.9	9.4.10	9.4.11	9.4.12	9.4.13	9.4.14	9.4.15	9.4.16
9.4.17	9.4.18	9.4.19	9.4.20	9.4.21	9.4.22	9.5.1	9.5.2	9.5.3	9.5.4	9.5.5	9.5.6	9.5.7	9.5.8	9.5.9	9.5.10
9.5.11	9.6.1	9.6.2	9.6.3	9.6.4	9.6.5	9.6.6	9.6.7	9.6.8	9.6.9	9.6.10	9.6.11	9.6.12	9.6.13	9.6.14	9.6.15
9.6.16	9.6.17	9.6.18	9.6.19	9.6.20	9.6.21	9.6.22	9.6.23	9.6.24	9.7.1	9.7.2	9.7.3	9.7.4	9.7.5	9.7.6	9.7.7
9.7.8	9.7.9														

FUNCTION - Controlling the Operation of the Ship and Care for Persons On Board at the Operational Level

10. COMPETENCE - Application of leadership and teamworking skills

10.1.1	10.1.2	10.1.3	10.1.4	10.1.5	10.1.6	10.1.7	10.2.1	10.2.2	10.2.3	10.2.4	10.2.5

11. COMPETENCE - Ensure compliance with pollution prevention requirements

11.1.1	11.1.2	11.1.3	11.1.4	11.1.5	11.1.6	11.1.7	11.1.8	11.1.9	11.1.10	11.1.11	11.2.1	11.2.2	11.2.3	11.2.4	11.3.1
11.3.2	11.4.1	11.4.2	11.5.1	11.5.2	11.6.1	11.6.2	11.6.3	11.6.4	11.6.5						

12. COMPETENCE - Maintain seaworthiness of the ship

12.1.1	12.1.2	12.1.3	12.1.4	12.1.5	12.1.6	12.1.7	12.1.8	12.1.9	12.1.10	12.1.11	12.1.12	12.2.1	12.3.1	12.3.2

13. COMPETENCE - Prevent, control and fight fires on board

13.1.1	13.1.2	13.1.3	13.1.4	13.1.5	13.2.1	13.2.2	13.3.1	13.3.2	13.4.1	13.4.2	13.4.3	13.4.4	13.4.5	13.4.6	13.4.7
13.4.8	13.4.9	13.4.10	13.4.11	13.5.1	13.5.2	13.5.3	13.5.4	13.5.5	13.5.6	13.6.1	13.6.2	13.6.3			

14. COMPETENCE - Operate life-saving appliances

14.1.1	14.1.2	14.1.3	14.1.4	14.1.5	14.1.6	14.1.7	14.1.8	14.1.9	14.2.1	14.2.2	14.2.3	14.2.4	14.2.5	14.2.6	14.2.7
14.2.8	14.3.1	14.4.1	14.5.1	14.5.2	14.5.3	14.5.4	14.5.5	14.5.6	14.5.7	14.5.8	14.5.9	14.5.10	14.5.11	14.5.12	

15. COMPETENCE - Apply medical first aid on board ship

15.1.1	15.1.2	15.1.3	15.1.4	15.2.1	15.2.2	15.3.1	15.3.2	15.3.3	15.3.4	15.3.5

16. COMPETENCE - Monitor compliance with legislative requirements

16.1.1	16.1.2	16.1.3	16.1.4	16.1.5	16.2.1	16.2.2	16.2.3	16.2.4	16.3.1

NB The STCW competence 'Contribute to the safety of personnel and ship' has been incorporated into competences 12-16





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